One Flow, Two Notions of Quality: Sampling Budgets for Image Inpainting

Supplementary Analyses and Implementation Details

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A Training duration and task quality

We track task quality during training on 124 development images with paired large and fragmented masks matched for missing area, spatial coverage, and foreground occlusion. FM and MSE Regression share the 14,986,435-parameter CNN, initialization within each seed, image and mask stream, and 30k-update schedule. Results cover seeds 0 and 1. FM uses eight function evaluations (FM8), while MSE Regression uses one (Regression1).

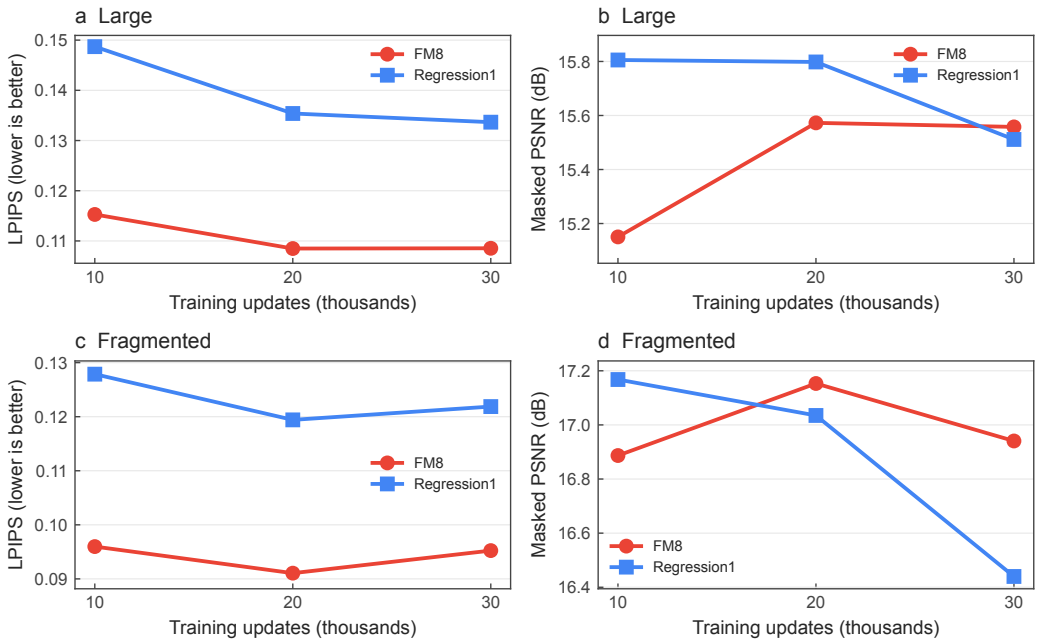


Figure S1: Development quality at 10k, 20k, and 30k using the same first mask pair per image. LPIPS covers the composed image and PSNR the missing region. FM scores average four fixed noise samples. Curves show means over two training seeds.

Most FM gains occur between 10k and 20k (Figure S1), followed by some deterioration toward 30k. For fragmented masks, mean LPIPS rises from 0.09106 at 20k to 0.09523 at 30k, while PSNR falls from 17.153 to 16.941 dB. Regression also exhibits a tradeoff between perceptual similarity and pixel fidelity across checkpoints. These trajectories show why task metrics are useful alongside the training objective. Intermediate checkpoints characterize learning under the shared 30k training budget.

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B Objective diagnostics and FM time bins

To examine how optimization relates to these task-quality trajectories, we evaluate fixed probes on 128 training images and 124 development images under both geometries. FM probes cover five equal time bins, whose average gives the overall FM objective. The online and exponential-moving-average (EMA) networks use the same probes.

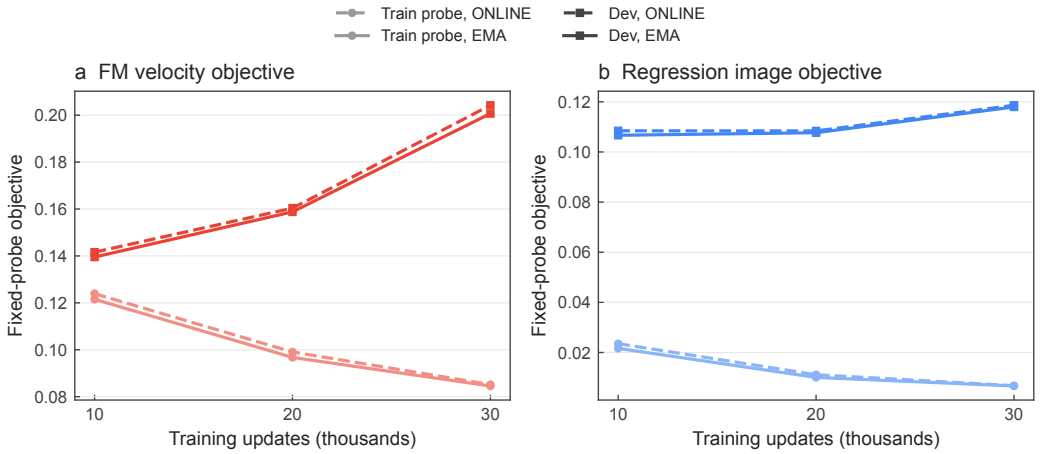


Figure S2: Fixed-probe objectives averaged over mask geometries and two seeds. Dashed lines show online networks and solid lines EMA networks. FM velocity MSE and regression image MSE have method-specific targets and scales.

Training-probe objectives decrease while development-probe objectives increase over the recorded checkpoints (Figure S2). The widening gap is consistent with overfitting under these objectives, but task quality follows a different trajectory. In particular, FM task quality initially improves despite the rising development velocity loss, and LPIPS and PSNR favor different checkpoints (Figure S1). Objective values and completion metrics therefore provide complementary views of training.

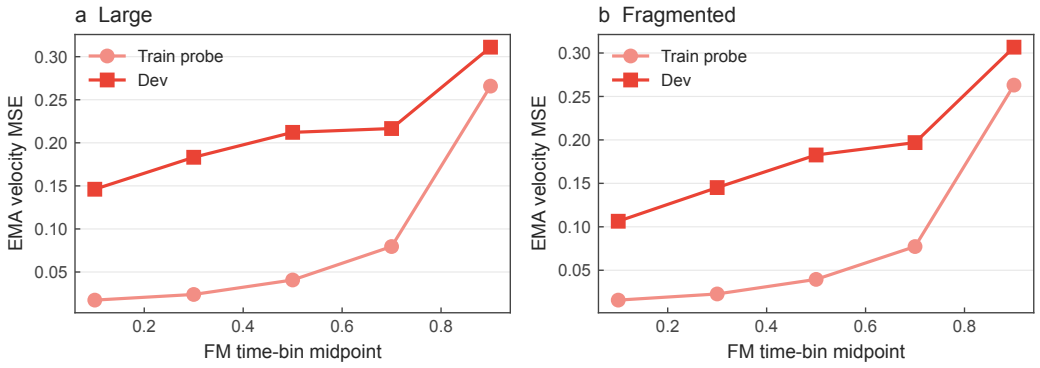


Figure S3: EMA FM velocity MSE at 30k across five equal time bins, averaged over two seeds. Points summarize fixed probes within each bin, and lines connect the bin summaries.

Several time bins show a marked gap between training and development losses, revealing how the overall objective gap is distributed along the interpolation path.

C CNN capacity at a fixed update budget

We explore CNN capacity within FM by comparing three widths using seed0, the same matched masks and supported images, 30k updates, and FM8. Final development metrics average four matched-mask pairs per image and four fixed noise samples per condition. This exploration informs the CNN width used in the subsequent method comparison.

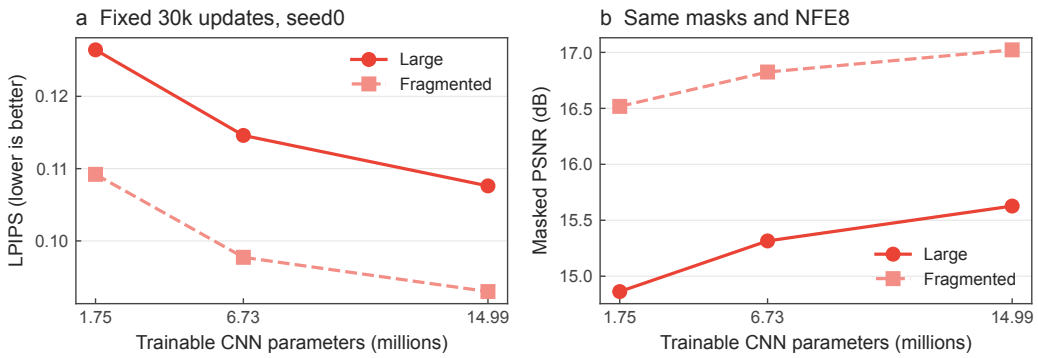


Figure S4: Development quality across three CNN widths at 30k updates and FM8. Each width uses one training seed (seed 0).

Table S1: Development quality averaged equally over both mask geometries, using seed 0, 30k updates, and FM8.

CNN parameters	LPIPS ↓	Masked PSNR (dB) ↑
1,747,523	0.11782	15.690
6,734,723	0.10617	16.070
14,986,435	0.10030	16.325

Table S2: Resources for the width comparison on the 48 GB RTX 4090. Train time excludes checkpoint evaluation, while total time includes it. Peak allocation is the maximum logged PyTorch allocation, excluding other CUDA allocations.

Parameters (M)	Train (min)	Total (min)	Peak allocation (GiB)
1.75	20.34	21.30	1.60
6.73	21.10	22.57	2.59
14.99	30.89	33.43	4.28

The largest tested width achieves the best development quality and is used for the subsequent two-seed comparison. This choice increases training time and memory as well as parameter count. The comparison thus measures quality at a fixed update budget, with the resource table showing the additional compute required by each width.

D Sampling computation on the development set

With CNN capacity and training duration fixed, we evaluate the final 14.99M FM checkpoints with 1, 2, 4, 8, 16, and 32 function evaluations. Each checkpoint uses the same development images, four matched-mask pairs per image, and four fixed noise samples. Regression1 provides a deterministic reference on the same masks. This development-set sweep isolates the effect of sampling computation within each trained FM model.

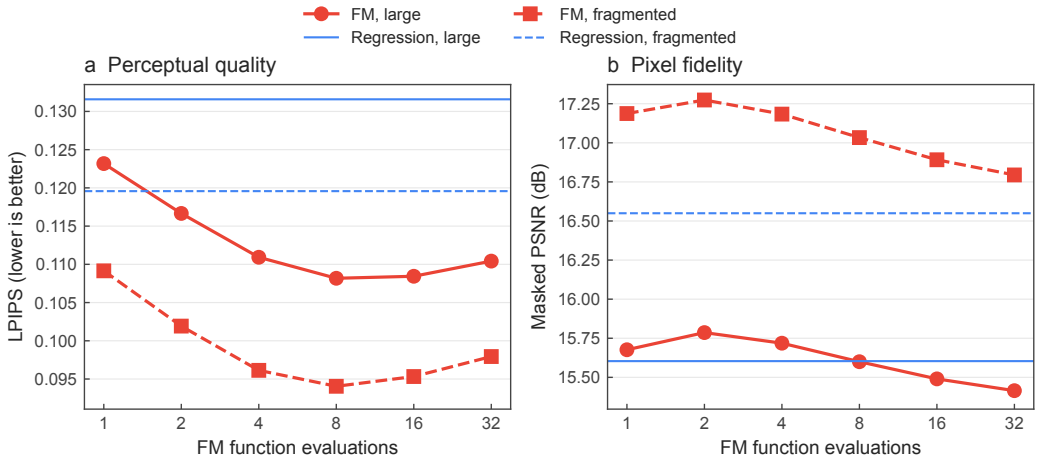


Figure S5: Development quality versus NFE. Red curves show means over seeds 0 and 1; blue horizontal lines show the mean pixel-regression reference for each geometry. Solid and dashed lines distinguish large and fragmented masks.

Both seeds and both geometries attain their lowest LPIPS among the tested NFE values at eight evaluations, whereas PSNR peaks at two evaluations. Further sampling therefore changes the balance between perceptual similarity and pixel fidelity before both metrics deteriorate at larger budgets. The shared locations of the best sampled values also indicate that the two geometries favor the same budgets in this comparison. The official test protocol evaluates the previously fixed NFE8 setting, with the full sweep confined to development images.

E Experimental provenance and reporting

Figures S1–S5 are rebuilt from existing development artifacts. The capacity comparison uses seed0 from runs v4, v5, and v6. The 14.99M learning and NFE analyses use the FM and Regression runs v6–v8 with seeds 0 and 1. Earlier v3 runs used a different mask design and are excluded.

Archived development tables are provided in `research_code/results/development/`. The ZIP also includes research code, protocol files, and test summaries. The code package README describes the included files and reproduction procedures. Figures S1–S5 cover FM and MSE Regression. The perceptual-loss Regression control was designed after the first official-test results were observed, and its development and test results are reported separately in the main paper.

The inherited near-duplicate review examined perceptual-hash distances up to four, and GeoStep repeats exact resized-pixel checks. These checks do not guarantee animal-identity separation. Both development and official-test images were used in an earlier classification project, while GeoStep excludes test images from training and configuration selection.

Timing uses GPU-resident BF16 prediction, five warm-ups, and 20 synchronized repetitions for one completion per image. I/O and metrics are excluded. Batch-64 times report amortized throughput.

F Implementation and mask-matching details

The experiments use PyTorch 2.7.1+cu128 with BF16 autocasting on the recorded RTX 4090 system. All trainable CNNs are initialized from scratch. The only pretrained network used for training is the frozen ImageNet VGG16 feature extractor in perceptual regression. AdamW uses its default moment coefficients $(0.9, 0.999)$ and numerical stability constant 10^{-8}, weight decay 10^{-4}, batch size 64, and gradient-norm clipping at 1.0. The EMA decay is 0.995. Each update samples image indices and mask-pair indices with replacement. Alternating batch positions select the two geometries, giving 32 conditions of each type. Images and masks are jointly flipped with probability 0.5.

The exact shared learning-rate recipe for update k is:

$$\eta_k = \begin{cases} \min(1, k/100)\left[4\times10^{-5} + 8\times10^{-5}\big(1+\cos(\pi k/2000)\big)\right], & 1 \le k \le 2000, \\ 4\times10^{-5}, & 2000 < k \le 10000, \\ 4\times10^{-5} + 4\times10^{-5}(k-10000)/1000, & 10000 < k \le 11000, \\ 10^{-5} + 3.5\times10^{-5}\left[1+\cos\left(\pi(k-11000)/19000\right)\right], & 11000 < k \le 30000. \end{cases} \tag{1}$$

All final comparisons use the fixed 30k EMA endpoint.

The seven-channel CNN input concatenates the current state, the visible image, and the mask. Residual blocks have two 3×3 convolutions, GroupNorm with eight groups, and SiLU activations. A projected time embedding is added after the first convolution, and the residual sum is scaled by $1/\sqrt{2}$. Time uses 32 logarithmically spaced frequencies, yielding 64 sine/cosine features, followed by a two-layer 128-dimensional MLP. The encoder has three stride-two 3×3 convolutions. The decoder uses nearest-neighbor interpolation, 3×3 convolutions, and concatenated encoder skips. The restoration CNN is trained from scratch and uses no attention layers.

For matching, trimaps define core foreground, possible foreground, and uncertain pixels. The allowed difference in missing core/possible foreground pixels is $\min(32, \lfloor 0.03A \rfloor)$, where A is the corresponding annotated region area. Available head boxes use the same rule, and uncertain-region counts differ by at most 64 pixels. Missing area, quadrant counts, and the actual central 16×16 mask pattern are identical within each pair. Candidate selection samples among the 32 eligible candidates with the smallest summed normalized foreground/head count differences. No model outputs enter mask selection. Train retains 2,812 of 2,924 offered images, development 124 of 128, and test 3,585 of 3,669. Head annotations are available for 2,805 retained training images, all retained development images, and no test images in the local annotation files. Missing head annotations supply no additional head-specific constraint.

G Perceptual regression and evaluation details

The VGG feature loss uses relu1_2, relu2_2, and relu3_3. It averages L_1 distances over channels and spatial dimensions within each layer, then equally over the three layers. Before optimization, 64 supported training images give masked image MSE 0.29171923 and feature loss 0.86761689, yielding scale $c = 0.33623046$. The two tested coefficients αc are 0.03362305 and 0.33623046, with the scale fixed for both seeds. Alpha selection uses seed0's final development LPIPS, with higher PSNR and then lower alpha as tie-breakers. Only the selected alpha is repeated for seed1. Perceptual-regression test inference follows this selection and checkpoint freeze.

The evaluator uses LPIPS package 0.1.4 with SqueezeNet and calibration version 0.1. LPIPS inputs are clipped to $[-1, 1]$ and evaluated at native resolution on full composed images. For PSNR, the clipped completion and reference are mapped to $[0, 1]$ before computing MSE on the missing pixels, using a peak intensity of 1. Each FM completion is scored independently before averaging over its four fixed noise samples and mask pairs, so reported PSNR is the mean of individual logarithmic scores. Bootstrap intervals resample original images, preserving all within-image pairings and the two-seed average. The test comparison uses 10,000 resamples. These intervals condition on the two trained seeds and describe image-level uncertainty rather than variation over a population of training runs.